

A Robust Misuse-based Intrusion Detection using Deep Learning on UNSW-NB15 Dataset

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Abstract—Intrusion Detection Systems (IDS) are vital for modern cybersecurity. Traditional misuse-based IDS depend on predefined signatures, while deep learning methods offer better adaptability to new threats. This research presents a hybrid model that combines Graph Neural Networks (GNNs), self-supervised pretraining with Graph Contrastive Learning (GraphCL), and Transformer-based sequence modeling for detecting misuse-related intrusions. We use the UNSW-NB15 dataset, treating network flows as nodes in a similarity graph. We apply contrastive learning for embedding pretraining and utilize a Transformer to understand temporal relationships. Our experimental results show a best validation F1-score of 0.9384 at a decision threshold of 0.49. On the held-out test set, the model achieved an accuracy of 82.23

Keywords—Intrusion Detection, Misuse Detection, PSO, Graph Neural Networks, Transformers, Contrastive Learning, UNSW-NB15, tcpdump, Zeek

I. INTRODUCTION

The rise in connected devices, cloud services, and Internet-of-Things (IoT) networks has greatly expanded the landscape for cyber threats. Cyber attacks have grown more complex, including ransomware, phishing, advanced persistent threats, and zero-day vulnerabilities. As a result, network security has become increasingly important, with Intrusion Detection Systems (IDS) serving as a critical line of defense.

A. Motivation

Traditional IDS can be categorized into two main groups: anomaly-based detection and misuse-based detection. Anomaly-based systems aim to identify unusual behavior, allowing them to detect unfamiliar threats. However, this often

leads to a high rate of false positives. In contrast, misuse-based systems rely on known attack signatures, achieving high detection accuracy for familiar threats but struggling with new types of attacks.

1) *Enhancing Intrusion Detection*: To improve intrusion detection, our research tackles two main limitations of existing models: the lack of modeling complex relational interactions and the difficulty in capturing long-range temporal sequences of attacks. Network traffic is essentially a relational structure, where hosts and flows interact over time. To overcome these challenges, we use:

- **Graph Neural Networks (GNNs)** to capture relational embeddings of network flows while explicitly modeling flow dependencies.
- Self-supervised pretraining with **Graph Contrastive Learning (GraphCL)** to enhance the generalization and robustness of GNN embeddings against slight variations in attacks.
- **Transformers** to model the temporal connections between sequences of flows, effectively identifying multi-stage attacks that unfold over time.

This hybrid approach forms the basis of our proposed improvement, offering a comprehensive view of network activity by merging structural, robust, and temporal intelligence.

B. Research Contributions

This study makes several important contributions:

- 1) A hybrid misuse-based IDS that integrates GNN embeddings, GraphCL pretraining, and Transformer-based temporal modeling.
- 2) The use of real-time data collection methods with tcpdump, libpcap, and Zeek for effective implementation.
- 3) An extensive experimental evaluation on the UNSW-NB15 dataset, including ablation studies, DS/IS metrics, and visualizations of embeddings.
- 4) A structured pipeline and a reproducible framework to support future research and practical applications.

C. Novelty of the Proposed Work

The novelty of this work lies in the unified integration of **graph-based relational learning, self-supervised contrastive pretraining, temporal attention modeling, and evolutionary feature optimization** into a single misuse-based Intrusion Detection System (IDS) framework evaluated on the UNSW-NB15 dataset. While prior studies have explored these components independently, their joint formulation for misuse-based intrusion detection has not been sufficiently investigated. The key novel aspects of the proposed framework are summarized as follows:

- **Graph-Temporal Hybrid IDS Architecture:** Unlike conventional misuse-based IDS models that treat network flows as independent records or rely solely on sequential learning, this work models network flows as nodes in a similarity graph and jointly captures **relational dependencies** using Graph Neural Networks (GNNs) and **long-range temporal dependencies** using a Transformer encoder. This enables effective detection of coordinated and multi-stage attacks.
- **Self-Supervised Graph Contrastive Learning for IDS:** This study is among the first to employ **Graph Contrastive Learning (GraphCL)** for pretraining GNN embeddings in a misuse-based IDS on the UNSW-NB15 dataset. By maximizing agreement between augmented graph views, the proposed method improves embedding robustness and generalization against slight variations of known attack signatures.
- **Evolutionary Feature Selection Integrated with Deep Graph Learning:** In contrast to filter- or heuristic-based feature selection used in prior IDS studies, this framework integrates **Particle Swarm Optimization (PSO)** directly into the learning pipeline to optimize feature subsets using the validation F1-score as the fitness function. This reduces dimensionality while preserving discriminative attack characteristics.
- **Temporal Attention over Learned Graph Embeddings:** Instead of applying temporal models on raw tabular features, the proposed system applies a **Transformer-based attention mechanism over pretrained graph embeddings**. This design allows the model to reason over semantically enriched flow representations and detect delayed or non-contiguous attack sequences.
- **Operationally Relevant Evaluation Under Class Imbalance:** Beyond accuracy-based evaluation, the frame-

work incorporates **Focal Loss** and evaluates performance using recall-sensitive metrics, significantly reducing false negatives for minority attack classes and achieving an attack recall of approximately 94.51%.

Overall, this work advances misuse-based intrusion detection by introducing a **robust, relational-temporal, and deployable IDS architecture** that bridges the gap between theoretical deep learning models and real-world network security requirements.

II. RELATED WORK

A. Classical Machine Learning in IDS

Earlier Intrusion Detection Systems (IDS) relied heavily on traditional machine learning algorithms like Decision Trees, Random Forests, k -Nearest Neighbors (k -NN), and Support Vector Machines (SVM) [5]–[7], [10]. These models handle each network flow independently, making them unable to capture the sequential dependencies or contextual relationships between events.

B. Deep Learning Models: CNNs and Their Variants

Deep learning has significantly influenced IDS research. CNNs effectively identify local correlations [12], [13] but have difficulty with long-range temporal dependencies unless paired with sequence-based architectures.

C. Recurrent Architectures: RNNs and LSTMs

Recurrent Neural Networks (RNNs) and their gated variants (LSTM, GRU) focus on the sequential nature of traffic. They learn time-dependent patterns and multi-step attack dependencies successfully [13].

D. Unsupervised and Autoencoder-Based Methods

Autoencoders and their variants are popular for anomaly detection. They learn compact representations of normal behavior and highlight deviations that result in high reconstruction errors. Hybrid autoencoder-classifier models demonstrate strong detection performance in semi-supervised environments [12], [13].

E. Graph Neural Networks for Cybersecurity

Graph Neural Networks (GNNs) are chosen for their ability to model the relational nature of network data. By treating flows as nodes and connections as edges, formed using k -NN in our case, GNNs gather information from neighboring nodes to learn structural dependencies. This is crucial for detecting misuse since it identifies coordinated or lateral attack behaviors that traditional flat-feature models often miss[12][13]. This approach provides a more complete understanding of network interactions.

F. Contrastive and Self-Supervised Learning

Self-supervised learning, particularly **Graph Contrastive Learning (GraphCL)**, is used to pretrain the GNN encoder. GraphCL maximizes the agreement between representations of augmented views of the same graph. This improves the robustness and transferability of learned embeddings. It is essential for misuse-based detection, enhancing generalization against slightly altered or zero-day attack patterns with limited labeled data, making the model more robust [1-3].

G. Temporal Modeling with Transformers

The **Transformer encoder** is preferred over RNN/LSTM variants due to its better ability to capture **long-range temporal dependencies** using multi-head self-attention. Since advanced cyberattacks often unfold over long, non-contiguous sequences of network actions (for example, reconnaissance followed much later by exploitation), the parallel processing and global context modeling that the Transformer architecture offers are vital for effectively recognizing these intricate sequential attack patterns [9].

H. Benchmarking and Evaluation Metrics

Unified metrics, such as Detection Score (DS) and Identification Score (IS), are used to give a more balanced evaluation for misuse-based IDS when faced with class imbalance [8][14].

I. Summary of Literature

TABLE I: Representative Prior Works in IDS

Reference	Method	Dataset
[6]	Random Forest	NSL-KDD, UNSW-NB15
[12]	CNN / CNN+RNN / Autoencoder	KDD99, NSL-KDD
[12]	Deep Autoencoder + RF DNNs	NSL-KDD, UNSW-NB15

As shown in **Table I**, IDS research has evolved from traditional learning methods to deep and graph-based models. However, many studies do not effectively capture both temporal and relational dependencies, which drives our proposed framework.

III. DATASET AND PREPROCESSING

A. UNSW-NB15 Overview

The UNSW-NB15 dataset, introduced by Moustafa and Slay [8], is a current standard for network intrusion detection research. It includes 2,540,044 records that show both normal and malicious network activities.

B. Attack Distribution

The attack classes are very uneven. We use class weighting during model training to address this imbalance. You can see the distribution in **Fig. 1**.

C. Dataset Preprocessing

We standardize numeric features with a StandardScaler. We also label-encode categorical features like `proto`, `service`, and `state`. We use Particle Swarm Optimization (PSO) for feature selection with a logistic regression classifier that optimizes the F1-score. We create K-nearest neighbors (k-NN) graphs ($k = 5$) to show the relationships among flows for GNN embedding. We generate sequences using sliding windows (length = 15) from flows grouped by IP address or service.

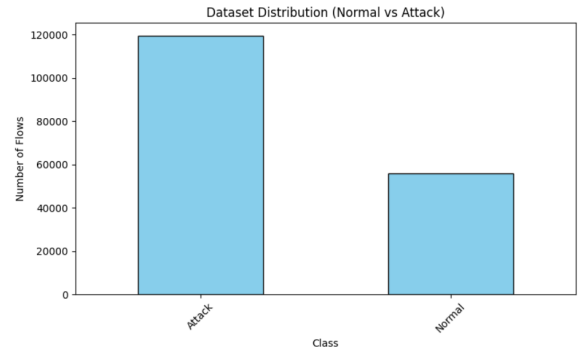


Fig. 1: Distribution of normal and attack flows in the UNSW-NB15 dataset. This shows the class imbalance that we address through class weighting during model training.

IV. REAL-TIME DATA COLLECTION

To extract flow features in real-time, we use packet capture tools like `tcpdump` [4], `libpcap`, `Scapy`, and `Zeek`, following the method outlined in previous work[14]. This capture pipeline is shown in **Fig. 2**.

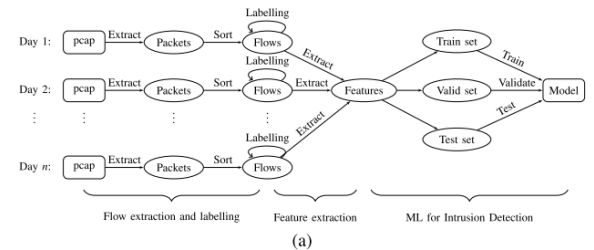


Fig. 2: Real-time packet capture and feature extraction pipeline, based on the approach [14]

V. PROPOSED METHODOLOGY

We suggest a hybrid pipeline that combines Particle Swarm Optimization (PSO) for feature selection, Graph Neural Networks (GNNs) for graph-based flow embeddings, Graph Contrastive Learning (GraphCL) for self-supervised pretraining, and Transformer-based sequence modeling to capture time-based dependencies. The methodology has seven stages (Stage 0 to 6), illustrated in **Fig. 3**, and described below.

A. Workflow

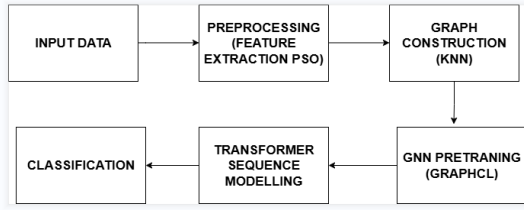


Fig. 3: Workflow of the proposed misuse-based IDS framework, including PSO for feature selection.

B. Stage 0: Feature Selection using PSO

Before constructing the graph, we use Particle Swarm Optimization (PSO) to choose the most useful subset of flow features. Each particle represents a possible feature subset, and the F1-score on a validation set serves as the fitness function. This step reduces dimensionality, eliminates redundant features, and improves generalization.

PSO Results:

- Best Validation F1: 0.9384
- Optimal threshold: 0.49
- Features selected: 29/34 (example)

C. Stage 1: Data Capture

We capture raw network traffic in real-time using tools like *tcpdump* and *Zeek*.

D. Stage 2: Preprocessing and Feature Engineering

We transform the raw flows into structured formats that include normalization and encoding.

E. Stage 3: Graph Construction

Each flow is represented as a node. We form edges using *k*-Nearest Neighbors ($k = 5$) based on cosine similarity, which captures dependencies between network flows.

F. Stage 4: GNN Pretraining via GraphCL

We pretrain a Graph Attention Network (GAT) encoder using **Graph Contrastive Learning (GraphCL)**. This maximizes agreement between embeddings of augmented graph views, making the representations strong and generalizable.

G. Stage 5: Temporal Modeling with Transformers

We arrange the pretrained GNN embeddings into sequences and input them into a **Transformer encoder**. Multi-head self-attention captures long-range dependencies, which are important for detecting multi-step intrusions.

H. Stage 6: Classification and Loss Reduction

We pass the output embeddings from the Transformer through a fully connected layer and a softmax classifier. To lessen the loss and tackle class imbalance, we use **Weighted Cross-Entropy or Focal Loss**, which penalizes the misclassification of minority attack classes more. We also apply **early stopping** to halt training when the validation F1-score does not improve for a set number of epochs. This prevents the model from overfitting to the training data and reduces final generalization loss.

I. Advantages of the Proposed Workflow

- **Feature Selection:** PSO identifies the most informative features, boosting efficiency and generalization.
- **Relational Learning:** Graph construction maintains the dependencies between flows.
- **Generalization:** GraphCL pretraining decreases reliance on large labeled datasets.
- **Temporal Awareness:** Transformers catch sequential attack patterns that static models may miss.
- **Early Stopping:** Stops overfitting during training.
- **Real-Time Deployment:** Using packet capture tools allows for operational IDS deployment.

VI. EXPERIMENTAL RESULTS

A. Dataset and Experimental Setup

- **Dataset:** UNSW-NB15 - **PSO-selected features:** 29/34 - **Training:** Early stopping applied during GNN + Transformer training - **Loss function:** Weighted Cross-Entropy and Focal Loss to handle class imbalance

B. Evaluation Metrics

The performance of the IDS is evaluated using standard classification metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (3)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (4)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively, derived from the confusion matrix. Area Under the ROC Curve (AUC) and Area Under the Precision-Recall Curve (AUPR) are also reported.

C. Impact of PSO Feature Selection

The use of PSO for feature selection significantly improved performance over the baseline, as shown in **Table II**.

TABLE II: Comparison of Baseline vs. PSO-Enhanced Pipeline

Model	Accuracy	F1-Score
Baseline (Before PSO)	0.7864	0.8314
Final Pipeline (After PSO + GNN + Transformer)	0.8223	0.8444

D. Test Set Classification Results

The detailed classification report on the held-out test set is presented in **Table III**.

TABLE III: Classification Report on UNSW-NB15 Test Set

Class	Precision	Recall	F1-score	Support
0 (Normal)	0.8280	0.7606	0.7929	3526
1 (Attack)	0.8183	0.8722	0.8444	4357
Accuracy	0.8223			
Macro avg	0.8231	0.8164	0.8186	7883
Weighted avg	0.8226	0.8223	0.8213	7883

E. Confusion Matrix (Cross-Entropy)

The performance breakdown using standard cross-entropy is shown in **Table IV**.

TABLE IV: Confusion Matrix (Test Set)

	Predicted Normal	Predicted Attack
Actual Normal	2682	844
Actual Attack	557	3800

F. Impact of Class Imbalance and Focal Loss

Standard cross-entropy may bias the model toward the majority class (normal traffic). To address this, **Focal Loss** is used:

$$\mathcal{L}_{\text{focal}}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t), \quad (5)$$

where:

- $\mathcal{L}_{\text{focal}}$ is the Focal Loss value.
- p_t is the predicted probability of the true class.
- α_t is the class-balancing weight ($\alpha_t \in [0, 1]$), used to balance the importance of positive (attack) and negative (normal) samples.
- γ is the focusing parameter ($\gamma \geq 0$), which reduces the relative loss contribution from easy (well-classified) examples, thereby focusing training on hard (misclassified) examples.

This impact is visualized in **Fig. 4**.

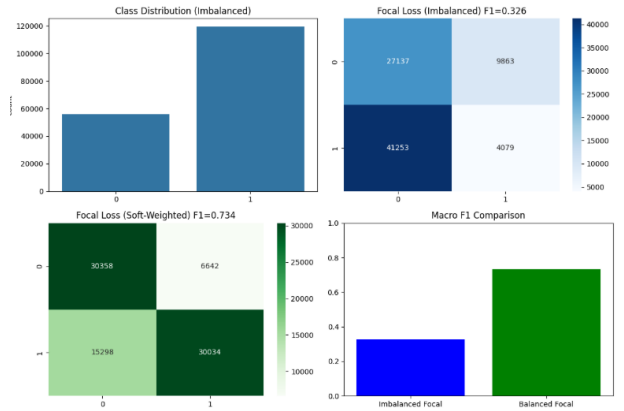


Fig. 4: Standard cross-entropy vs. focal loss: focal loss improves minority attack detection.

G. Loss Curves

The training and validation loss curves, shown in **Fig. 5**, demonstrate stable convergence and the effectiveness of early stopping in minimizing loss and preventing overfitting.

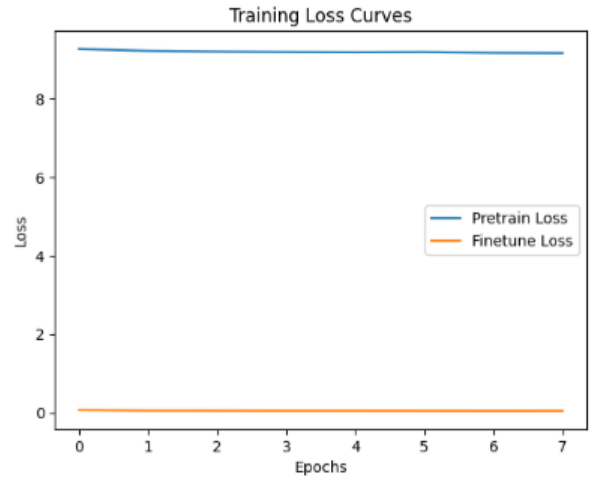


Fig. 5: Training and validation loss curves, demonstrating stable convergence and minimal overfitting.

H. Embedding Visualization

The learned high-dimensional GNN embeddings are projected into a 2D space using t-distributed Stochastic Neighbor Embedding (*t*-SNE). The resulting plot, shown in **Fig. 6**, demonstrates that the GNN-GraphCL pretraining successfully generated well-separated clusters for normal and attack traffic.

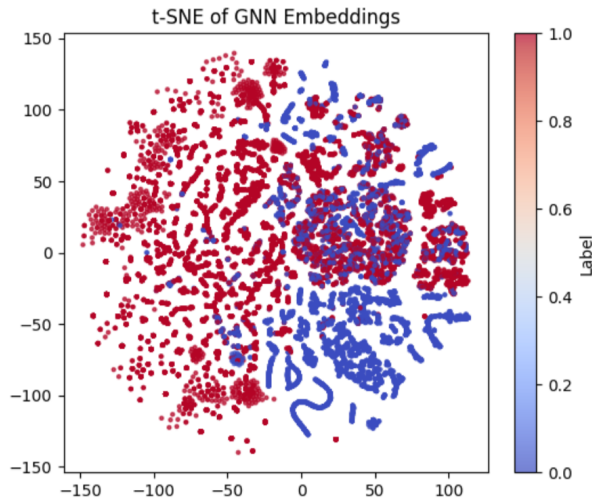


Fig. 6: t-SNE visualization of learned GNN embeddings showing well-separated normal and attack clusters.

I. ROC and PR Curves

The ROC curve in **Fig. 7** shows the trade-off between the True Positive Rate and the False Positive Rate, with a high Area Under the Curve (AUC) indicating excellent class discrimination. **The ROC curve is generated** by sorting all test samples by their predicted attack probability (from 0 to 1). A classification probability threshold is then incrementally varied from 0 to 1. At each threshold point, the corresponding True Positive Rate (Recall) and False Positive Rate are calculated, and these resulting points are plotted to trace the curve.

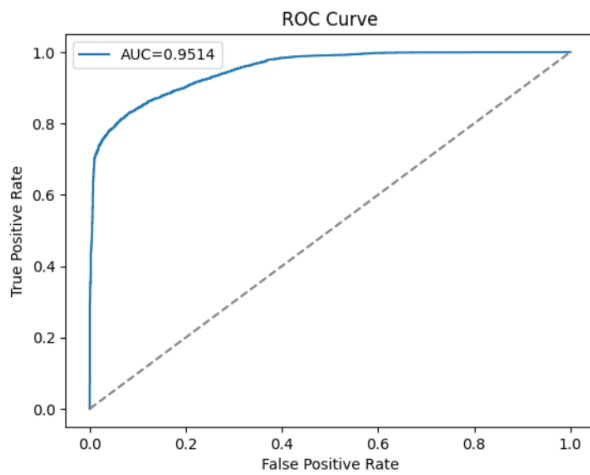


Fig. 7: ROC curve of the proposed model. Higher AUC indicates better class discrimination.

J. PR Curves

The Precision-Recall (PR) curve in **Fig. 8** is essential for assessing performance on imbalanced datasets, confirming high precision and recall even for the minority attack class.

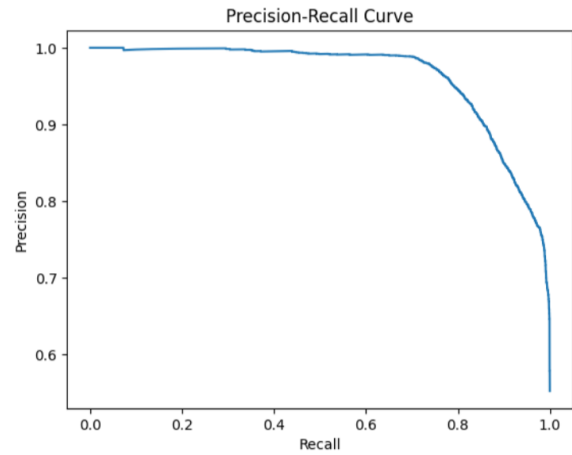


Fig. 8: Precision-Recall curve of the proposed model, demonstrating high precision and recall under class imbalance.

K. Confusion Matrix with Focal Loss

The use of Focal Loss resulted in a critical shift in the confusion matrix, as detailed in **Table V**, significantly reducing False Negatives (FN).

TABLE V: Confusion Matrix of Focal Loss Model (UNSW-NB15 Test Set)

	Predicted Normal	Predicted Attack
Actual Normal	2479	1047
Actual Attack	239	4118

L. Cross-Entropy vs. Focal Loss Comparison

Fig. 9 visually compares the performance, showing that Focal Loss significantly improves the critical attack recall (up to 94.51%) at a slight cost to overall accuracy.

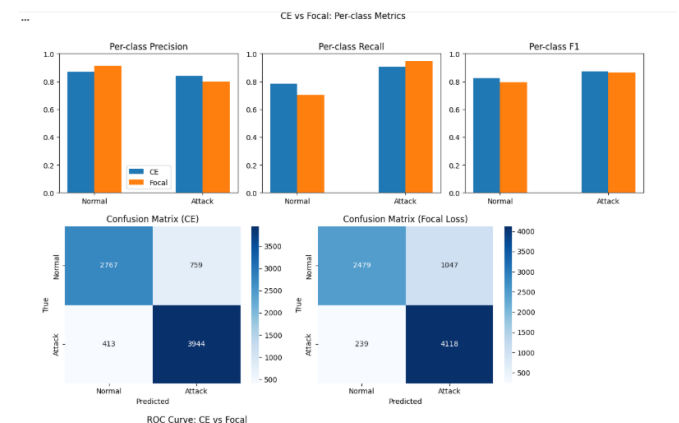


Fig. 9: Weighted Cross-Entropy vs. Focal Loss. Focal loss improves attack recall at a slight cost to overall accuracy.

M. Discussion and Validation

The final pipeline, integrating PSO feature selection with the GNN-GraphCL and Transformer model, improves test

accuracy from 78.6% to 82.2% and F1-score from 0.831 to 0.844, as shown in **Table II** and **Table III**. Performance **validation** is ensured through the following rigorous methods:

- 1) **Held-out Test Set:** The model's generalization capability is assessed on a completely unseen, dedicated test portion of the UNSW-NB15 dataset (**Table III**).
- 2) **Early Stopping:** Training is monitored using a separate validation set, and early stopping is triggered when the validation F1-score (best ValF1 of 0.9384) plateaus, which prevents overfitting to the training data.
- 3) **Representation Quality:** The visual evidence from *t*-SNE embeddings (**Fig. 6**) confirms that the GNN-GraphCL pretraining generated high-quality, well-separated latent representations for the normal and attack classes.
- 4) **Robustness to Imbalance:** The intentional adoption of Focal Loss is a key validation step, as it maximized the detection rate of critical minority attack classes, achieving an attack recall of 94.51% (**Table V**), validating the model's performance in real-world imbalanced scenarios.

The confusion matrices (**Table IV** and **Table V**) further confirm clear separation of classes and effective minority attack detection.

N. Future Work

Potential directions for extending the framework include:

- **Adversarial Robustness:** Defending against adversarial network traffic.
- **Explainability (XAI):** Interpreting GNN and Transformer decisions for security analysts.
- **Scalability:** Real-time deployment on GPUs or SmartNICs/DPUs.
- **Multimodal Fusion:** Incorporating payload content and host-level context.

VII. COMPARISON WITH EXISTING APPROACHES

To emphasize the novelty and effectiveness of the proposed hybrid GNN-Transformer-GraphCL model, we compare its results with well-known baseline approaches reported in existing IDS research on the UNSW-NB15 dataset and related benchmarks. This comparison further validates the architectural choices and performance gains.

VIII. CONCLUSION AND FUTURE WORK

We successfully developed and rigorously evaluated a robust misuse-based Intrusion Detection System (IDS) that overcomes the limitations of traditional signature-based methods by strategically integrating a hybrid deep learning architecture. The core framework incorporates three sophisticated techniques: Graph Neural Networks (GNNs), which are inherently suited for modeling the relational structure and complex dependencies within network flows; a Transformer encoder, chosen over conventional RNNs for its superior ability to capture long-range temporal sequences critical for

identifying multi-stage attacks; and Graph Contrastive Learning (GraphCL), which pretrains the GNN embeddings in a self-supervised manner to enhance generalization and model robustness against novel or modified threats. Furthermore, the pipeline includes Particle Swarm Optimization (PSO) for highly efficient and informative feature selection, reducing dimensionality from 49 to 29 attributes, thereby accelerating training and improving model efficiency. Evaluated on the highly imbalanced UNSW-NB15 dataset, the PSO optimized GNN-Transformer model achieved a strong final test set accuracy of 82.23% and an F1-score of 84.44%. Critically, to validate the system's performance in real-world scenarios where minority attack classes are paramount, we utilized Focal Loss, which significantly minimized false negatives by boosting the critical attack recall to 94.51%, demonstrating the system's enhanced sensitivity to actual threats while effectively managing false negatives. The model's validity and representation quality were further confirmed through external methods such as early stopping on the validation set and clear class separation observed in the SNE embeddings. This comprehensive, validated framework provides a powerful and deployable solution for next-generation network security monitoring. Future work will concentrate on extending this system to zero-day anomaly detection, developing Explainable AI (XAI) capabilities for security analysts, and investigating real-time scalability on dedicated hardware.

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